

Smit Chaudhary

QUANTUM ALGORITHMS DEVELOPER · QUANTUM PHYSICIST · SOFTWARE ENGINEER

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Experience

Quantum Algorithms Developer

Amsterdam, The Netherlands

PASQAL

Nov 2022 - Present

- Developed Quantum Scientific Machine Learning (QSciML) algorithms. Built and trained variational quantum circuits to solve industrially relevant aerodynamics PDEs, creating quantum alternatives to fully classical physics-informed neural networks (PINNs)
- Served as technical lead on European Union funded projects, translating quantum research into practical solutions and driving business development opportunities with major aerospace industry partners
- Architected quantum software infrastructure and built hardware-aware circuit capabilities, making strategic technical decisions that aligned algorithms with neutral atom hardware constraints while enhancing both open-source and proprietary frameworks
- Mentored 3 research interns and 2 junior developers, leading to successful project contributions and 3 research publications

Quantum Computing Intern

California, USA

MENTEN AI

May 2022 - Sept 2022

- Developed a novel fully quantum generative adversarial network (QGAN) architecture for discrete and binary data generation
- Investigated and implemented quantum circuit features such as noise reuploading in the generator and auxiliary qubits in the discriminator to enhance expressivity
- Demonstrated the model's performance on synthetic datasets and low-energy states of Ising models, showing successful data reproduction and generalization capabilities

Education

Technische Universiteit Delft

Delft, The Netherlands

MASTER OF SCIENCE IN APPLIED PHYSICS

August 2020 - July 2022

- **Thesis:** Quantum Walk Based Algorithms for Qubit Mapping (**Supervisor:** Prof. Sebastian Feld)
- **Research Experience:** Honors Project on Noise Induced Barren Plateaus at Leiden University (**Supervisor:** Prof. Jordi Tura)
- **Relevant Coursework:** Applied Quantum Algorithms, Quantum Information, Quantum Computing Architecture

Indian Institute of Technology, Kanpur

Kanpur, India

BACHELOR OF SCIENCE IN PHYSICS

July 2016 - May 2020

- **Undergraduate Project:** Bohmian Mechanics and Quantum Information (**Supervisor:** Prof. Kaushik Bhattacharya)
- **Research Experience:** Summer Research at IISER Kolkata on Quantum Machine Learning (**Supervisor:** Prof. Prasanta Panigrahi)
- **Relevant Coursework:** Quantum Computing, Quantum Field Theory, Statistical Mechanics, Probability and Statistics, Optics

Selected Publications

Solving Fluid Dynamics Equations with Differentiable Quantum Circuits

S. Chaudhary, G. T. Balducci, O. Kyriienko, P. K. Barkoutsos, L. Cardarelli, A. A. Gentile

Proceedings of the 35th Parallel CFD International Conference 2024, May 2025

Quantum Circuit Training with Growth-Based Architectures

C. Duffy, S. Chaudhary, G. V. Velikova

arXiv preprint, Nov 2024

Towards a scalable discrete quantum generative adversarial neural network

S. Chaudhary, P. Huembeli, I. MacCormack, T. L. Patti, J. Kossaifi, A. Galda

Quantum Science and Technology, April 2023

Skills

Programming

Python, C/C++, Rust, Verilog

Languages

English, Hindi, Gujarati

Libraries & Utilities

Git, PyTorch, NumPy, PennyLane, Tensorflow, Jax, Cirq, Qiskit